

2 Subject content and assessment information

Qualification aims and objectives

The aims and objectives of this qualification are to encourage students to:

- understand the application of techniques within the framework of the statistical enquiry cycle and the research methodologies used in experiments and surveys
- apply statistical techniques to data sourced from a variety of contexts, appreciating when samples or population data could be used and applying appropriate sampling techniques
- generate and interpret the diagrams, graphs and measurement techniques used in performing statistical investigations
- have an understanding of how visualisations of multivariate data are used to gain a qualitative understanding of the multiple factors that interact in real life situations, including, but not limited to, population characteristics, environmental considerations, production variables etc.
- understand how technology has enabled the collection, visualisation and analysis of large data sets to inform decision-making processes in public, commercial and academic sectors
- develop skills in interpretation and critical evaluation of methodology including justifying the techniques used for statistical problem solving
- apply appropriate statistical formulae, as set out in *Appendices 1* and *2*.

Content

Appendix 1 lists the statistical formulae that students are expected to know and recall.

Appendix 2 lists the statistical formulae and tables that will be available to students during the examination.

Appendix 4 contains information on the expectations relating to the use of calculators and knowledge of language and techniques associated with the use of spreadsheets and databases.

1 Numerical measures, graphs and diagrams

Subject content	Additional information
1.1 Interpret statistical diagrams including bar charts, stem and leaf diagrams, box and whisker plots, cumulative frequency diagrams, histograms (with either equal or unequal class intervals), time series and scatter diagrams.	Students will not be required to draw or construct statistical diagrams. Students may be required to comment on or interpret the main features of published visualisations (for example those listed opposite) or critically assess published visualisations. This may include examples which draw upon multivariate data.*
1.2 Know the features needed to ensure an appropriate representation of data using the above diagrams, and how misrepresentation may occur.	
1.3 Justify appropriate graphical representation and comment on those published.	
1.4 Compare different data sets, using appropriate diagrams or calculated measures of central tendency and spread: mean, median, mode, range, interquartile range, percentiles, variance and standard deviation.	Students may be required to comment on or interpret the main features of published visualisations (for example those listed in section 1.1). This may include examples which draw upon multivariate data.*
1.5 Calculate measures using calculators and manual calculation as appropriate.	Where raw data is given candidates will be expected to obtain the values of the mean, standard deviation and variance directly from a calculator.
1.6 Identify outliers by inspection and using appropriate calculations.	
1.7 Determine the nature of outliers in reference to the population and original data collection process.	
1.8 Appreciate that data can be misrepresented when used out of context or through misleading visualisation.	This may include examples which draw upon multivariate data.*

*Please see *Appendix 4: Use of calculators and other technology*.